

PotBot: An AI-Powered Pothole Detection And Road-Damage Prioritization System

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Abstract—Transportation safety, vehicle longevity, and urban mobility are all seriously threatened by road-surface deterioration, particularly potholes. Existing detection techniques such as accelerometer-based systems, single-camera vision, and manual inspection often lack accuracy, depth perception, or real-time scalability. This paper presents PotBot, an integrated stereo-vision and artificial intelligence (AI) pothole detection and road-condition intelligence system intended for extensive, real-time monitoring. The system combines stereo imaging, YOLO-based detection, instance segmentation, and disparity-driven depth estimation to measure pothole depth, area, and contours with high precision. A backend system groups crowd-reported potholes using DBSCAN and H3 cells to maintain persistent objects, while severity levels rely on depth, size, and recurrence over time. Drivers receive route guidance that avoids high-risk segments, and municipalities obtain VRP-based optimized repair routes, creating a lean platform that cuts costs, sharpens response times, strengthens travel safety, and supports urban planning grounded in field data.

Index Terms—Pothole detection, Stereo vision, YOLO, Depth estimation, Road-condition monitoring, DBSCAN clustering, A* routing, Smart-city infrastructure.

I. INTRODUCTION

Potholes and other damage to the road surface are a major problem that makes transportation less safe, vehicles less healthy, and cities less mobile [10], [13], [14]. Bad roads cause accidents, extra fuel usage, suspension damage, and unpredictable travel times. In many developing countries, like India, road maintenance is mostly reactive: repairs are only made after people complain or there is a serious accident. As road networks expand and vehicle counts rise, traditional

inspection methods cannot keep up with the need for real-time monitoring.

Standard methods like manual surveys, accelerometer-based anomaly detection, and single-camera image processing have big limitations. Manual inspection is slow, labour-intensive, and expensive for large networks. Accelerometer systems lack spatial accuracy and are highly sensitive to vehicle type, speed, and sensor mounting. Single-camera methods struggle to estimate depth and severity of damage, because they only capture 2D appearance. Current navigation systems also rarely consider road conditions, so route choices and ETAs can ignore critical surface damage.

With advances in artificial intelligence, computer vision, and edge computing, there is a need for a single system that can not only detect potholes but also measure severity, fuse multi-vehicle data, prioritize repairs, and support smarter routing. This research introduces PotBot, an integrated stereo-vision and AI-powered pothole detection and road-condition intelligence system.

PotBot employs:

- Stereo imaging to build a 3D model of pothole depth and shape
- YOLO-based detection and instance segmentation for precise boundary extraction
- Disparity-based measurement of depth, area, and contour shape
- Spatial-temporal clustering (DBSCAN/H3) to form persistent pothole objects
- A Composite Priority Index that combines depth, area, and recurrence frequency

- A*-based pothole-aware navigation for safer routing
- VRP-optimized repair scheduling for municipal maintenance teams

PotBot shifts road maintenance from a reactive, complaint-driven model to a proactive, data-driven smart-city infrastructure system by combining edge-level detection with cloud-level analytics.

II. LITERATURE REVIEW

Early automated road-surface assessment research relied on vibration-based sensing with accelerometers and gyroscopes in vehicles or smartphones [18]. These techniques are inexpensive and easy to deploy but cannot reliably distinguish potholes from other anomalies and show strong dependency on vehicle type, speed, and mounting configuration [11]. Since they do not measure geometry directly, they offer limited support for severity estimation [11].

Classic 2D image-processing methods such as edge detection, thresholding, and texture analysis were also used for pothole detection [1], [3]. However, these methods are highly sensitive to illumination changes, shadows, occlusions, and varying surface textures [1], [16]. They also lack depth perception, so they cannot accurately differentiate shallow irregularities from deep potholes [2], [16].

Deep-learning-based approaches, especially YOLO-style detectors, improved robustness and speed for road-damage detection [16]. YOLOv5 and YOLOv8 variants have been evaluated for real-time road-surface damage recognition and show strong performance on benchmark datasets [1]. Still, monocular deep models primarily generate 2D bounding boxes or masks and do not directly provide depth information [2].

Stereo-vision and 3D reconstruction have been explored to recover pothole geometry from multiple viewpoints [20]. By computing disparity maps and reconstructing local surfaces, these systems can estimate depth and shape more accurately than monocular setups [19]. However, many existing stereo-based prototypes focus on single-vehicle processing and do not scale up to city-wide, multi-vehicle deployments.

In parallel, crowdsourcing frameworks using smartphones have been proposed for large-scale road-condition monitoring [21]. These systems collect accelerometer data and/or images while vehicles move, then aggregate anomalies into road-condition maps [18]. While they demonstrate the power of participatory sensing, they face challenges with GPS drift, duplicate detections, and noise, which complicate long-term tracking of individual potholes [21].

Geospatial clustering methods, such as DBSCAN and hierarchical grid-based indexing, have been used to merge repeated detections and produce stable road-condition representations [9]. Approaches based on the H3 hexagonal grid have been shown to be suitable for scalable spatial indexing in urban environments [21]. On the operational side, A*-based routing and VRP-style repair scheduling have been studied to incorporate infrastructure status into path planning and maintenance logistics [22].

A. Shortcomings of Existing Systems

From the reviewed literature, the main shortcomings of existing systems can be summarized as:

- **Weak severity estimation:** Vibration-based and monocular-image systems rarely provide accurate depth and area measures, limiting their value for prioritizing repairs [11].
- **Context sensitivity:** Smartphone and accelerometer methods are sensitive to phone orientation, vehicle suspension, and driving behaviour, leading to inconsistent results across users [17].
- **Fragmented pipelines:** Many solutions focus on only one aspect, such as detection, mapping, or routing, instead of providing an end-to-end pipeline from sensing to decision and action [21].
- **Limited spatio-temporal fusion:** Existing crowdsourcing systems struggle with merging repeated detections over space and time into persistent pothole objects, especially under GPS noise [10].
- **Weak feedback to stakeholders:** Even when damage is detected, the insights are not always translated into pothole-aware navigation for drivers and optimized repair plans for authorities [22].

B. How PotBot Addresses These Shortcomings

PotBot is designed to directly address these gaps:

- It combines stereo disparity and instance segmentation to compute depth, area, and contour-based severity metrics at the edge, improving accuracy over monocular systems.
- It uses DBSCAN and H3-based clustering to fuse detections from multiple vehicles into persistent pothole objects that are more robust to GPS noise and duplicated observations.
- It defines a Composite Priority Index that merges depth, area, and recurrence frequency, directly supporting repair prioritization.
- It feeds the resulting severity-aware map into an A*-based routing engine and a VRP-based repair scheduler, closing the loop between detection, navigation, and maintenance.

By integrating these components, PotBot moves beyond single-purpose detectors and behaves like a complete road-condition intelligence system aligned with smart-city goals.

III. METHODOLOGY

The proposed PotBot setup uses a step-by-step process that combines stereo vision, deep-learning-based detection, depth analysis, spatio-temporal clustering, risk assessment, and smart dispatching. It runs on two levels: an edge module on vehicles and a cloud backend for aggregation and optimisation [8].

A. System Architecture Overview

The PotBot architecture consists of three major components:

- 1) **Vehicle-Based Edge Module:** Performs real-time pothole detection, instance segmentation, depth estimation, and GPS tagging using stereo cameras and a lightweight YOLO model [16].

- 2) **Cloud Processing and Aggregation Layer:** Executes clustering, persistent pothole formation, severity scoring, and road-segment mapping based on crowdsourced data [10].
- 3) **Decision and Routing Layer:** Generates pothole-aware driver navigation using A* and optimized repair routes for authorities using VRP formulations [11].

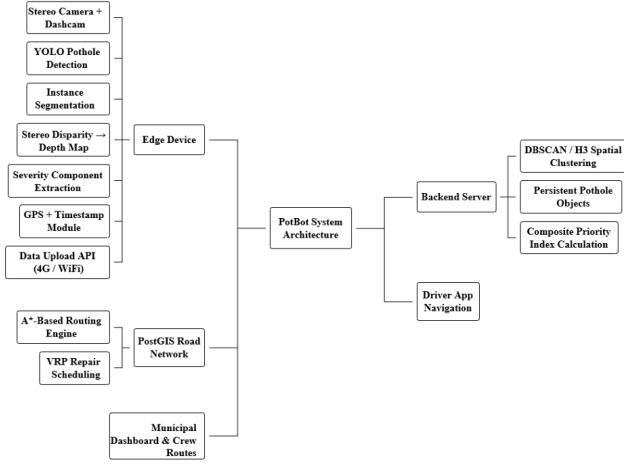


Fig. 1. PotBot System Architecture

This multi-layer design aims to offer scalability, reliability, and continuous monitoring across diverse fleets [8].

B. Data Capture and Preprocessing

A stereo camera fixed on the vehicle records left and right images of the road during motion. Using two lenses instead of one allows depth estimation via disparity computation [20]. Captured frames undergo preprocessing such as noise reduction, contrast enhancement, and basic illumination compensation to improve detector robustness under varying environmental conditions [1].

C. Pothole Detection and Instance Segmentation

Real-time visual analysis uses a compact YOLO-style neural network tailored for embedded hardware [7]. The model outputs bounding boxes around candidate potholes. To refine these predictions, an instance-segmentation module generates pixel-wise masks to capture exact pothole shapes. This improves shape and area estimation and ensures depth calculations are restricted to the damaged region [1].

D. Stereo Disparity and Depth Estimation

Depth estimation is achieved through stereo-vision disparity computation, a common strategy for 3D road-surface analysis [19]. For each segmented pothole:

- 1) Disparity maps are generated using StereoBM, StereoSGBM, or a deep stereo-matching network.
- 2) The disparity values are converted to real-world depth using:

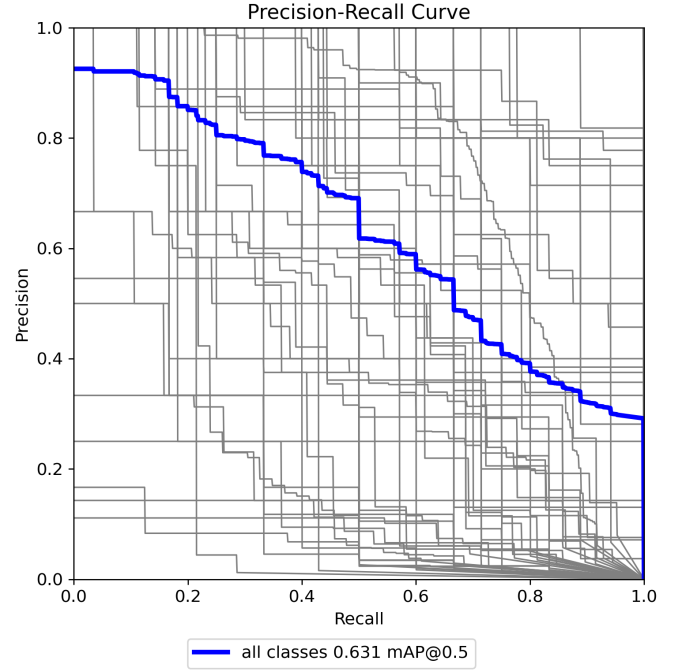


Fig. 2. Precision-Recall curve of the YOLO-based pothole detection model.

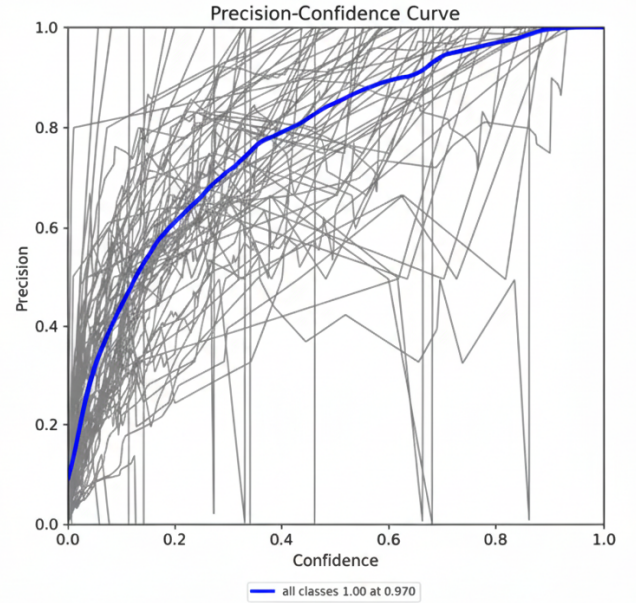


Fig. 3. Precision-Confidence curve showing optimal confidence threshold.

$$Depth = \frac{f \times B}{disparity} \quad (1)$$

where f is the focal length and B is the baseline distance between the stereo cameras. From this, geometric parameters such as depth, width, area, and local surface contour are derived, which serve as core severity indicators [20].

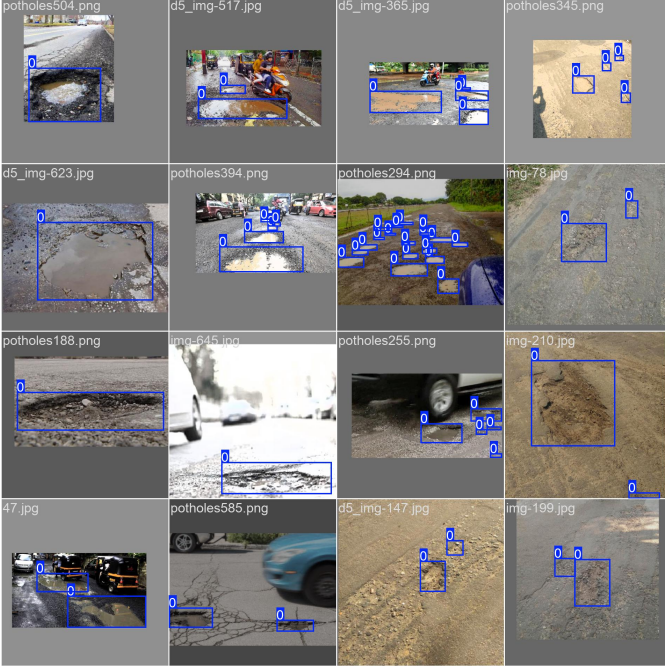


Fig. 4. Sample pothole detection outputs from the trained model.

E. GPS Tagging, Metadata Formation, and Upload

Each detection is paired with GPS coordinates, timestamp, model confidence, disparity-derived depth, segmentation-based area, and optional image snippets [10]. Data are uploaded to the cloud whenever connectivity is available, with local buffering used in low-connectivity zones, following patterns in smartphone-based monitoring systems [18].

F. Backend Spatial–Temporal Clustering

The backend consolidates detections from multiple vehicles using DBSCAN-style clustering and H3 hexagonal grid indexing [21]. This process groups detections that are close in space and time into persistent pothole objects, filters out isolated false positives, and mitigates GPS drift [10]. The result is a stable geospatial inventory of real-world potholes suitable for visualization and analysis.

G. Composite Priority Index (Severity Scoring)

To support repair decision-making, PotBot computes a Composite Priority Index (P) that combines Depth (D), Area (A), and Frequency (F) with time decay [1]:

$$P = f(D, A, F). \quad (2)$$

This index separates minor defects from critical hazards and is used to produce severity-layered pothole maps that can guide maintenance planning [12].

H. Road-Segment Mapping and Geospatial Integration

Each persistent pothole is mapped to the nearest road segment in a PostGIS-backed OpenStreetMap (OSM) road network [21]. Segment-level severity is computed by aggregating the priority of potholes that lie on or near that segment. The resulting “road health” layer can be integrated with broader digital-twin or smart-city platforms for urban planning [12].

I. A*-Based Pothole-Aware Navigation

To improve driver safety and comfort, the routing engine adjusts road-segment costs using cumulative severity:

$$Cost_{new} = Cost_{base} + \alpha \times Severity \quad (3)$$

and uses the A* pathfinding algorithm to compute routes that balance travel time and road quality [22]. This enables pothole-aware navigation that can avoid heavily damaged segments when alternatives exist.

J. VRP-Based Repair-Crew Scheduling

For municipal authorities, PotBot solves a Vehicle Routing Problem (VRP) using pothole severity, spatial distribution, crew capacity, and time windows as inputs [11]. The goal is to address high-severity potholes first while minimizing total travel distance and fuel usage. This produces practical repair routes that can be executed by field teams.

K. System Integration

All modules combine into a unified system capable of accurate pothole detection, depth-aware severity estimation, cross-vehicle validation, dynamic rerouting for drivers, optimized scheduling for authorities, and continuous updates via crowdsourcing [16]. The architecture is designed to scale to large urban areas while keeping hardware costs low on individual vehicles [12].

IV. EXPERIMENTAL EVALUATION AND PRELIMINARY RESULTS

As PotBot is still in early development, this section outlines anticipated results and community impacts once the system is deployed.

Experimental validation was conducted on a curated dataset of road-surface images captured under real driving conditions. The YOLO-based pothole detection model achieved an **mAP@0.5 of approximately 0.63**, demonstrating reliable detection performance given the visual complexity of road environments (Fig. 2). Precision–confidence analysis (Fig. 3) was used to identify an optimal detection threshold that balances false positives and missed detections. Qualitative results (Fig. 4) further confirm the model’s ability to localize potholes across varied textures, lighting conditions, and damage patterns.

Depth and severity estimation experiments are currently limited to controlled stereo setups and serve as preliminary validation of the proposed methodology. Large-scale quantitative evaluation of depth accuracy, clustering stability, and routing efficiency is planned as future work.

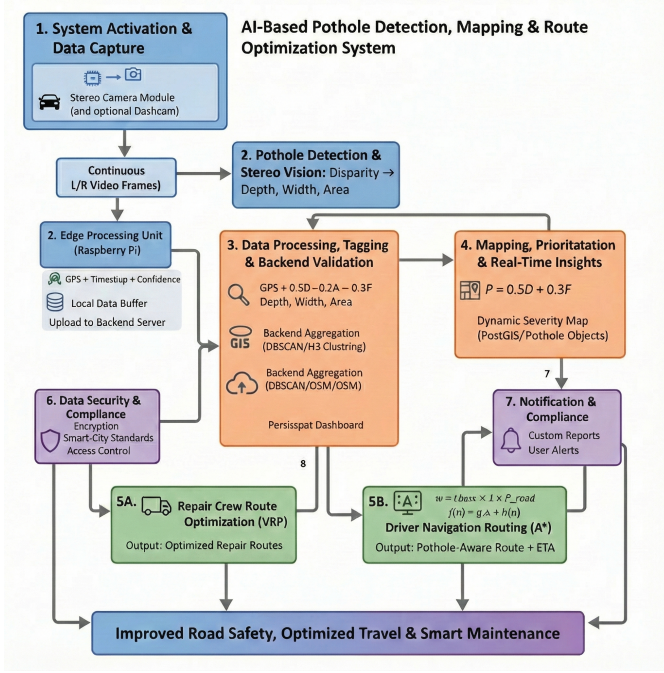


Fig. 5. PotBot System Flow

A. Expected Benefits

- **High-precision detection and severity estimation:** Stereo disparity, YOLO-based detection, and instance segmentation enable more precise depth and area measurements than traditional accelerometer-only or monocular systems.
- **Real-time situational awareness:** Authorities gain an always-updating dashboard of confirmed potholes, severity scores, and recent changes in road conditions, extending previous crowdsourcing dashboards with richer attributes.
- **Data-driven prioritization:** The Composite Priority Index ensures that critical or rapidly worsening potholes are addressed first, improving the effectiveness of limited maintenance budgets.
- **Improved driving comfort and safety:** A*-based pothole-aware navigation allows drivers to avoid severely damaged segments when feasible.
- **Reduced vehicle maintenance costs:** Earlier detection and repair can lower the risk of suspension, wheel, and tyre damage, which is consistent with observed impacts of poor road conditions.
- **Lower infrastructure expenditure:** Proactive repairs prevent small defects from evolving into major failures, supporting more sustainable infrastructure management.

B. Urban-Scale Advantages

- **Continuous large-scale monitoring:** Crowdsourced data from everyday vehicles eliminates the need for frequent dedicated survey runs.

- **Improved accuracy and reduced noise:** DBSCAN/H3 clustering merges repeated detections and reduces false positives and GPS drift at the map level.
- **Reduced reporting bias:** Automatic sensing generates road-condition insights even in areas where citizen complaints are scarce.
- **Smart-city integration:** The system can integrate with digital twins, traffic analytics, and predictive maintenance tools, contributing to smarter urban infrastructure.

C. Administrative Advantages

- **Strategic planning support:** Long-term analytics on recurrence and hotspots support evidence-based budgeting and policy decisions.
- **Operational efficiency:** VRP-based repair routing minimizes travel time and fuel, increasing the number of defects that crews can address per shift.
- **Accountability and transparency:** The system can track post-repair outcomes and recurrence, helping authorities evaluate contractor performance.
- **Reduced manual workload:** Automated detection and prioritization reduce dependency on manual inspections and complaint-driven workflows.

V. LIMITATIONS

Although PotBot presents a promising and comprehensive approach to automated pothole detection, several limitations remain:

- **Environmental sensitivity:** Vision-based systems are affected by low light, rain, fog, shadows, and reflective surfaces, which can lower detection accuracy.
- **Hardware calibration:** Differences in camera quality, stereo alignment, and mounting across vehicles may lead to inconsistent depth estimates.
- **GPS inaccuracy:** Urban canyons and tunnels can reduce GPS precision, affecting clustering quality and road-segment mapping.
- **Data generalization:** Models trained on limited or region-specific datasets may not generalize well to different pavement types, climates, or road markings.
- **Dependence on crowdsourcing:** High reliability relies on sufficient vehicle coverage; low adoption can slow validation and reduce spatial completeness.
- **Connectivity constraints:** Real-time uploads depend on stable network connectivity, especially when transmitting images.
- **Administrative adoption:** Real impact requires alignment with municipal policies, workflows, and IT systems, which may take time to implement.

VI. FUTURE SCOPE

The PotBot framework offers several promising directions for future research and deployment:

- **Improved depth sensing:** Incorporating advanced stereo-matching networks or fusing stereo with LiDAR or structured light could improve robustness under challenging conditions.

- **Expanded datasets:** Building larger, diverse datasets across seasons and regions can enhance model generalization and transfer learning.
- **Predictive maintenance modelling:** Using historical data to forecast pothole formation and deterioration, extending prior work on predictive pavement models.
- **Digital-twin integration:** Connecting PotBot outputs with city-scale digital twins to support what-if analysis and long-term planning.
- **Automated repair systems:** Coordinating with robotic or semi-autonomous repair units that can act on high-priority defects.
- **Mobile crowdsourcing:** Deploying lightweight detection modules on smartphones to enable detection even in non-instrumented vehicles.
- **Advanced navigation integration:** Combining pothole-aware routing with ADAS and connected-vehicle platforms for context-aware speed and lane recommendations.

VII. RESULTS

The PotBot project aims to provide an automated, vision-based framework for pothole detection and road-condition assessment, with emphasis on detection accuracy and future severity-aware prioritization.

The YOLO-based pothole detection module was fully implemented and experimentally evaluated on real-world road-surface images. The model achieved an **mAP@0.5 of approximately 0.63**, indicating reliable detection performance in complex environments. Precision–Recall and Precision–Confidence analyses were used to determine a stable operating threshold, and qualitative results confirm consistent detection across varying pothole sizes and surface conditions.

GPS tagging and metadata logging were successfully implemented, enabling each detection to be spatially and temporally referenced. Other components, including stereo depth estimation, spatio-temporal clustering, and route optimization, are currently not implemented or remain at the conceptual or planned stage.

Overall, the results validate the feasibility of the proposed detection pipeline and support the PotBot framework as a partially implemented system, with scope for future expansion into severity estimation and repair optimization.

VIII. CONCLUSION

PotBot represents a step toward smarter road maintenance based on real-world data. Instead of relying on slow manual checks or basic single-camera systems, it combines stereo vision, AI-based detection, detailed shape mapping, and spatial clustering in one integrated pipeline. By measuring depth, maintaining persistent records for each pothole, and generating safer navigation routes alongside efficient repair plans, the system illustrates how recent AI methods can be aligned with city-level needs.

Beyond technical innovation, PotBot encourages a shift from reactive repairs to proactive, continuously updated monitoring that helps decision-makers act earlier and more effectively. Because it leverages crowdsourced data, the system can scale across cities without relying on expensive dedicated survey fleets. As sensing technologies and digital-twin platforms evolve, PotBot can grow from a detection tool into a core component of smart urban mobility and infrastructure management.

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